The most accurate model was looking at the wrong thing

PneumoScan · project summary · live demo pneumoscan-nine.vercel.app · code github.com/Prathamesh250205/PneumoScan-CNN

Three ImageNet-pretrained CNNs were fine-tuned to detect pneumonia in pediatric chest X-rays (Kermany et al., 5,856 images). ResNet18 scored highest, but only 44% of its Grad-CAM attention fell on the lungs, barely above the 39% expected by chance. MobileNetV2 was 2.2 points less accurate and put 71% there. On 656 adult X-rays from another hospital, every model lost about 0.2 AUC and ResNet18's lead disappeared.

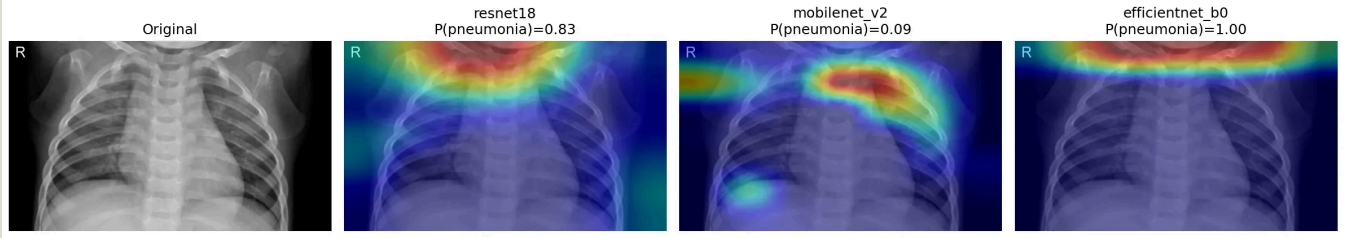


Fig. 1 A normal X-ray. ResNet18 predicts pneumonia (P = 0.83) with attention on the neck and upper mediastinum. MobileNetV2 correctly predicts normal (P = 0.09) with attention on the lung.

1 · In-distribution performance

Model	Accuracy	F1	AUC (95% CI)	Sens.	Spec.	Spec. at 95% sens.*
ResNet18 (served)	87.8%	0.911	0.978 (0.967–0.987)	0.997	0.679	0.882
MobileNetV2 (served)	85.6%	0.896	0.968 (0.953–0.980)	0.995	0.624	0.863
EfficientNet-B0	84.8%	0.891	0.951 (0.933–0.968)	0.997	0.598	0.790

Kermany test split, n = 624 (234 normal, 390 pneumonia), threshold 0.5 unless noted. *Threshold chosen for 95% sensitivity by stratified 2-fold cross-validation (50 splits). For ResNet18 this is t = 0.946, used by the live demo.

2 · Where the models look

Lungs, clavicles and scapulae were segmented with torchxrayvision's ChestX-Det PSPNet. For each image we measured the share of Grad-CAM energy (last conv block, predicted class) inside the lung mask. Normal X-rays are the clean comparison, because the adult-trained segmenter drops some consolidated lung on pediatric pneumonia X-rays.

Normal X-rays	Pediatric: on lungs (95% CI)	Peak in lung	Adult NIH: on lungs (95% CI)
chance (lung share of image)	*38.5%*		*33.4%*
ResNet18	43.9% (42.8–45.0)	21%	38.7% (36.8–40.6)
MobileNetV2	**71.2%** (70.1–72.3)	**60%**	**58.2%** (56.3–60.0)
EfficientNet-B0	40.7% (38.1–43.4)	33%	16.4% (14.1–19.0)

MobileNetV2 − ResNet18: **+27.3 points** on pediatric (CI +26.0 to +28.6) and +19.6 on adult X-rays. A correction to this project's first, visual claim: ResNet18's attention on the clavicles and scapulae (4.7%) is *below* their 6.4% share of the image. Its attention isn't on the shoulders. It's spread over the neck, mediastinum and borders, and mostly not on the lungs.

3 · External test: another hospital, adult patients

656 X-rays from NIH ChestX-ray14 (batch 7) with radiologist-adjudicated labels (Majkowska et al., 2020): 364 airspace opacity, 292 with all expert labels negative, mostly AP views.

Model	Pediatric AUC	NIH AUC (95% CI)	Drop (95% CI)
ResNet18	0.978	0.772 (0.735–0.808)	−0.206 (−0.246 to −0.170)
MobileNetV2	0.968	**0.786** (0.753–0.819)	**−0.181** (−0.217 to −0.146)
EfficientNet-B0	0.951	0.733 (0.695–0.771)	−0.218 (−0.259 to −0.178)

- ResNet18's pediatric lead is significant (paired bootstrap ΔAUC −0.011, CI −0.019 to −0.003 for MobileNetV2 − ResNet18) but vanishes externally. MobileNetV2 ranks first, though +0.015 (CI −0.020 to +0.051) is not significant.
- Thresholds don't transfer. At the pediatric 0.946, ResNet18 flags 80% of healthy adults (specificity 0.199).
- Lung-focused attention did not buy measurable robustness under this shift, which mixes a new hospital with a new age group.

Method and system

- **Training:** PyTorch, ImageNet weights, 2-class head, Adam 1e-4, 10 epochs, flips and ±10° rotation (Kaggle T4).
- **Preprocessing:** grayscale → 3 channels → 224^2 bilinear → ImageNet normalisation. Identical in training and serving.
- **Statistics:** stratified bootstrap (2,000 draws), paired across models on the same images.
- **Serving:** ONNX Runtime + FastAPI as a Vercel function. The ONNX graphs also output the class activation map, which equals Grad-CAM for global-average-pool heads (verified correlation 1.0), so live heatmaps need no backward pass.
- **Testing:** Playwright end-to-end test on desktop and mobile, run against local and production.

Limitations

- One training run per model, so 1–3 point gaps between models may be partly seed noise.
- The external test mixes a hospital shift with an age shift (children → adults).
- Lung masks come from a segmentation model, not radiologists.
- The validation split has 16 images, so operating points were cross-validated on the test set and are optimistic for new sites.

> **Research demonstration only. Not a medical device.** Not validated for clinical use and not cleared by any regulator. Do not use for diagnostic decisions.

Data: Kermany et al., *Cell* 2018 · Wang et al., CVPR 2017 (ChestX-ray14) · Majkowska et al., *Radiology* 2020 · Cohen et al., MIDL 2022 (TorchXRayVision).